

B. Tech Ist Semester Sessional Examination-I
Subject-Mathematics-II
Branch- CSE (Sec-A), Total Marks-30, Date-10.02.2025

Answer the following questions:

1. Solve (any three):

f) $\frac{dy}{dx} + y \cot x = 2 \cos x$

g) $\frac{dy}{dx} + \frac{y}{x} = x^2 : y = 1 \text{ when } x = 1$

h) $\frac{dy}{dx} + \frac{y}{x} = y^2$

i) $(y - 2x^3)dx - x(1 - xy)dy = 0$

j) $(x + 2y - 2)dx + (2x - y + 3)dy = 0$

2. Answer the following:

d) Show that the function $f(z) = \bar{z}$ is not analytic

e) Show that the function $u = \frac{1}{2} \log(x^2 + y^2)$ is harmonic

f) Show that the function $f(z) = xy + iy$ is everywhere continuous but is not analytic

5x3=15

5x3=15